An Explicit and Scene-Adapted Definition of Convex Self-Similarity Prior With Application to Unsupervised Sentinel-2 Super-Resolution

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***Abstract*— Sentinel-2 satellite, launched by the European Space Agency, plays a critical role in various Earth observation missions. However, the spatial resolutions of Sentinel-2 images are different across its spectral bands, including four bands with a resolution of 10 m, six bands with a resolution of 20 m, and three bands with a resolution of 60 m. To facilitate the effectiveness of analyzing these images, super-resolving of the low-/medium-resolution bands to a higher resolution is desired. As in any image restoration inverse problems, we exploit image self-similarity, a commonly observed property in natural images, which underlies the state-of-the-art techniques, e.g., in image denoising. However, the design of self-similarity priors in nondiagonal inverse problems is challenging; often, a denoiser based on self-similarity is plugged into the iterations of an algorithm, without a guarantee of convergence in general. In this article, for the first time, we introduce a convex and scene-adapted regularizer built explicitly on a self-similarity graph directly learned from the Sentinel-2 images. We then develop a fast algorithm, termed Sentinel-2 super-resolution via scene-adapted self-similarity (SSSS). We experimentally show the superiority of SSSS over four commonly observed scenes, indicating the potential usage of our convex self-similarity regularization in other imaging inverse problems.**

***Index Terms*— Convex optimization, scene-adapted self-similarity, self-similarity, Sentinel-2 satellite, super-resolution.**

I. Introduction

SATELLITE remote sensing images have found widespread applications in Earth surface observations, including agricultural monitoring, change detection, and geophysical variable estimation [1], [2]. Owing to hardware limitations of the imaging sensors, as well as other practical concerns such as atmospheric absorption and satisfactory signal-to-noise ratio (SNR), the spatial resolution of multiband systems varies across the spectral bands. In such a scenario, many systems adopt ground sampling distances (GSDs) varying across the spectral bands in order to make an efficient use of the system resources and the communication bandwidth. Some well-known multispectral multiresolution satellites include the Moderate Resolution Imaging Spectroradiometer (MODIS) [3], Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) [4], as well as the very recent, Sentinel-2 launched by the European Space Agency (ESA) [5]. To facilitate the effectiveness of analyzing these images, super-resolving the lower resolution bands to make their spatial resolution the same as those with the highest resolution ones is of paramount importance.

Without loss of generality, we will focus on the Sentinel-2 satellite, which involves four bands with a resolution of 10 m, six bands with a resolution of 20 m, and three bands with a resolution of 60 m. Our aim is to super-resolve its coarser bands (i.e., 20-/60-m bands) to achieve the highest spatial resolution (i.e., 10 m). This problem is different from, and actually more challenging than, the conventional pansharpening problem and the hyperspectral/multispectral image fusion problem [6]–[11], both of which consider scenarios in which the spectral range of high-resolution counterpart images (i.e., panchromatic image or multispectral image) significantly overlaps with the spectral range of coarser images [12]–[14], whereas the high-resolution bands (10 m) and the coarser bands (20/60 m) in Sentinel-2 imagery are almost nonoverlapping in their spectral ranges (see Table I). Therefore, some benchmark pansharpening/fusion methods, such as multiresolution analysis (MRA) [15] and component substitution (CS) [16], are not our preferred solutions for the Sentinel-2 super-resolution task, for which most of the coarser bands are disjoint from those high-resolution bands in their spectral ranges, as can be seen from Table I. Although what we pursued here is a fully unsupervised approach, we should also mention the convolutional neural network (CNN)-based methods [17], [18]. In [18], a deep residual network is designed for low sensitivity to important hyperparameters (e.g., the number of residual blocks) based on the assumption

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TABLE I
INFORMATION OF SPECTRAL BANDS FOR THE SENTINEL-2 SENSORS

Band	B1	B2	B3	B4	B5	B6	B7	B8	B8a	B9	B10	B11	B12
Central Wavelength (nm)	443	490	560	665	705	740	783	842	865	945	1380	1610	2190
Bandwidth (nm)	20	65	35	30	15	15	20	115	20	20	30	90	180
GSD (m)	60	10	10	10	20	20	20	10	20	60	60	20	20

of scale invariance (e.g., the mapping from 20 to 10 m is similar to that from 40 to 20 m), trained at reduced resolution, and then applied to sharpen Sentinel-2 images at full resolution. In [17], the assumption of scale invariance is also adopted in the network training stage, while the training data sets were collected globally over a wide range of geographical areas (so that the user has no need to retrain the network for a particular land cover type as claimed by the authors). Because of its global applicability, the Deep Sentinel-2 (DSen2) method [17] is considered as a representative method in the CNN category. We remark that such global applicability echoes one important feature of this article, namely the scene adaptivity of the regularization scheme, meaning that the proposed regularizer will automatically adapt to the input data.

Since hardware improvements are out of reach, we will attack Sentinel-2 super-resolution computationally. One category of methods is rooted in the kriging-based geostatistical analysis. Some representative methods include downscaling cokriging (DSCK) [19], [20] and kriging with external drift (KED) [21], where the former requires heavy convolution/deconvolution computation in the cross-semivariogram modeling, while the latter simplifies such computation at the cost of much larger size of kriging matrices (and requires computing the kriging weights for each pixel). As the state-of-the-art method (in terms of super-resolution performance) in this category, area-to-point regression kriging (ATPRK) [22] made a breakthrough in the computational aspect. ATPRK separately handles the regression part and the residual downscaling and computes the kriging weights only once, yielding a computationally efficient algorithm that well preserves the spectral properties for those coarse images [22].

Another category of methods considers the super-resolution task as an imaging inverse problem [23]. Such an inverse problem is clearly ill-posed and, therefore, requires reducing the number of variables, such as by rank reduction of the (target high-resolution) Sentinel-2 data matrix [24], or adding suitable regularization scheme to the formulated super-resolution criterion. Some most commonly seen regularizers in imaging sciences are the total variation (TV) for promoting piecewise smoothness and preserving edges [25], [26], polytope volume (PV) for yielding high-fidelity spectra [27]–[30], and ℓ_0-norm (or its convex relaxation, i.e., ℓ_1-norm) for encouraging practically meaningful solutions (e.g., sparse material distributions) [14], [31]. Besides several variations of TV, the edge-preserving regularizer may also be designed, by learning the spatial discontinuities from high-resolution bands and then transferring them to those coarser bands [32]; such an approach implicitly exploits the high correlations among different spectral bands, which somehow echoes the validity of the aforementioned rank reduction approach [24].

The state-of-the-art Sentinel-2 super-resolution method in this category, referred to as multiresolution sharpening approach (MUSA) [33], exploits image self-similarity, which asserts that, in a given image of the real world, it is very likely to find many similar patches at different locations and scales. Many multiband images, such as multispectral images, hyperspectral images, videos, and MRI sequences, also exhibit self-similarity in the 3-D domain, associated with the sequence of bands. This is to say that patches, which seen as vectors in the high-dimensional embedding space, exhibit highly structured configurations that may be exploited by nonlocal processing schemes. Since the denoising seminal work [34] in 2005, using nonlocal means (NL-means), image self-similarity has underlaid a series of impressive state-of-the-art contributions to imaging inverse problems. The NL-means algorithm replaces each patch with a weighted average of similar patches. The algorithms [35] and [36] sparsely code the image patches in a dictionary learned from the noisy data, compute the denoised patches from the codes, and average each pixel. The algorithms [37] and [38] group together similar patches from a noisy image into a 3-D stack, compute a 3-D transform, hard threshold the coefficients, invert the 3-D transform, and average at each pixel. In [39], a mixture of Gaussian density (MoG) is learned from natural image patches and a linear inverse problem (LIP) is solved via expected patch log-likelihood (EPLL), which computes an approximate maximum *a posteriori* (MAP) estimate. Finally, in [40], a hyperspectral and multispectral fusion problem is solved by learning an MoG from the high spatial resolution patches of the multispectral image, build a denoiser with it, and then solve the fusion imaging inverse problem by plugging [41] the derived denoiser in an instance of alternating direction method of multipliers (ADMM) solver [42].

Although self-similarity plays a critical role in recent advances of Sentinel-2 super-resolution, there is no explicit definition for this regularizer. This is the case of MUSA, which relies on plugging in other self-similarity regularized denoiser [e.g., block-matching and 3-D filtering (BM3D)] [43] when implementing optimization step corresponding to the so-called proximity operator [44] of the self-similarity regularizer. The use of plug-and-play strategy has two problems: 1) the exact structure of the implemented regularization is hard to compute, if not impossible and 2) there is no provable converge guarantee for such plug-and-play strategy.

In this article, for the first time, the self-similarity phenomenon is explicitly defined under the convex framework; note that convexity is highly desirable in terms of

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efficient optimization. It is designed as a convex function ϕ of the eigenimages[1] of the target image (i.e., the super-resolved image). This allows us to formulate the Sentinel-2 super-resolution problem as a convex optimization problem. On the other hand, we solve the formulation by employing the split augmented Lagrangian shrinkage algorithm (SALSA) [42], which is an instance of ADMM [45] and guarantees that the algorithm convergences. Remarkably, under our definition of ϕ, the self-similarity regularizer is scene-adapted. Specifically, a self-similarity graph is automatically learned from the input multispectral multiresolution images, and such scene-adapted graph defines self-similarity as a convex regularizer. This not only ensures convergence but also yields a computationally very efficient algorithm termed Sentinel-2 super-resolution via scene-adapted self-similarity (SSSS). We will evaluate SSSS on four different scenes, including crops, mountains, coastal line, and urban area. Experimental results illustrate the state-of-the-art performance of SSSS.

The remaining part of this article is organized as follows. In Section II, we rigorously formulate the Sentinel-2 super-resolution problem. In Section III, we explicitly define self-similarity as a convex regularizer and develop SSSS, a computationally efficient algorithm to solve the corresponding convex problem. In Section IV, a set of experimental results are used to illustrate the superior performance of SSSS, followed by conclusions drawn in Section V. Some proofs are collectively presented in the appendixes.

Some standard notations adopted in this article are collectively presented hereinafter. $\mathbb{R}^n$ and $\mathbb{R}^{m\times n}$ denote, respectively (resp.), the set of real-valued n-dimensional vectors and the set of $m \times n$ matrices. $\mathbb{C}^{m\times n}$ denotes the set of complex-valued $m \times n$ matrices. $\mathcal{I}_Z \triangleq \{1, \ldots, Z\}$ for any positive integer Z. Let $\boldsymbol{F}_n \in \mathbb{C}^{n\times n}$ be the discrete Fourier transform (DFT) matrix, whose (j,k)th entry is $[\boldsymbol{F}_n]_{j,k} \triangleq (1/\sqrt{n})\exp[-(2\pi\imath/n)(j-1)(k-1)]$ (here $\imath \triangleq \sqrt{-1}$). The symbols $\mathbf{1}$, $\mathbf{0}$, and $\boldsymbol{I}$ represent, resp., an all-one vector, an all-zero vector/matrix, and an identity matrix of appropriate dimension. $\boldsymbol{e}_k^{(r)}$ denotes the kth r-dimensional unit vector. $[\boldsymbol{v}]_{a:b} \triangleq (v_a, \ldots, v_b)$ denotes the subvector of a given vector $\boldsymbol{v} = (v_1, \ldots, v_n)$, where $1 \le a \le b \le n$. The Euclidean norm and the Frobenius norm are denoted, resp., by $\|\cdot\|_2$ and $\|\cdot\|_F$. $\mathrm{vec}(\cdot)$ is the vectorization operator, while $\mathrm{vec}_{m\times n}^{-1}(\boldsymbol{v})$ is the $m \times n$ matrix $\boldsymbol{M}$ satisfying $\mathrm{vec}(\boldsymbol{M}) = \boldsymbol{v}$. The symbol $\mathrm{rank}(\boldsymbol{M})$ denotes the rank of a matrix $\boldsymbol{M}$. $\otimes$ denotes the Kronecker product. For a set $\mathcal{S}$, $|\mathcal{S}|$ is the cardinality of $\mathcal{S}$. For a matrix $\boldsymbol{M}$, $|\boldsymbol{M}|$ returns a matrix the (i,j)th entry of which is the absolute value of the (i,j)th entry of $\boldsymbol{M}$. Finally, $\mathrm{diag}(\boldsymbol{M})$ is the vector the ith entry of which is given by the ith diagonal entry of a given square matrix $\boldsymbol{M}$.

II. Problem Formulation

Formulating the super-resolution of multispectral multiresolution images is tricky due to the different spatial resolutions of different spectral bands. Sentinel-2 images have $\ell = 12$ spectral bands, including $\ell_1 = 4$ high-resolution bands with GSD = 10 m, $\ell_2 = 6$ medium-resolution bands with GSD = 20 m, and $\ell_6 = 2$ low-resolution bands with GSD = 60 m; another 60-m band, containing the cirrus information, is for atmospheric correction and not considered herein [46]. Our aim is to super-resolve all the bands to have a 10-m spatial resolution, so their upsampling factors are $r_1 = 1$, $r_2 = 2$, and $r_6 = 6$. The target image $\boldsymbol{X} \in \mathbb{R}^{\ell\times n}$ (i.e., the super-resolved Sentinel-2 images, the ℓ band images of which correspond to the ℓ rows in $\boldsymbol{X}$) contains $n = r_i^2 n_i$ high-resolution pixels, where n_i is the number of pixels in the band the upsampling factor of which is r_i, $\forall i \in \{1, 2, 6\}$.

Let $\boldsymbol{Y}_b \in \mathbb{R}^{n_{i,1}\times n_{i,2}}$ $(n_{i,1}n_{i,2} = n_i)$ be the bth band of the acquired data $\boldsymbol{Y}$. For notational conciseness, we insert zeros to $\boldsymbol{Y}_b$ to make its number of pixels n, followed by vectorization, leading to the vectorized representation of the image in the bth band of $\boldsymbol{Y}$, i.e., $\boldsymbol{y}_b \triangleq \mathrm{vec}(\boldsymbol{Y}_b \otimes (\boldsymbol{e}_1^{(r_i)}(\boldsymbol{e}_1^{(r_i)})^T) \in \mathbb{R}^n$. Then, they are stacked to obtain the vector representation of the data set $\boldsymbol{y} = [\boldsymbol{y}_1^T, \ldots, \boldsymbol{y}_\ell^T]^T \in \mathbb{R}^{\ell n}$; without loss of generality, we assume that $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_4$ correspond to the high-resolution bands, $\boldsymbol{y}_5, \ldots, \boldsymbol{y}_{10}$ correspond to the medium-resolution bands, and $\boldsymbol{y}_{11}, \ldots, \boldsymbol{y}_{12}$ correspond to the low-resolution bands.

Accordingly, let $\boldsymbol{x}_b \in \mathbb{R}^n$ be the vectorized image of the bth band of the target image $\boldsymbol{X}$. Then, let $\boldsymbol{x} \triangleq [\boldsymbol{x}_1^T, \ldots, \boldsymbol{x}_\ell^T]^T \in \mathbb{R}^{\ell n}$ be the vectorized representation of the target image $\boldsymbol{X}$. Then, the forward model can be described as [33]

$$\boldsymbol{y} = \boldsymbol{M}\boldsymbol{B}\boldsymbol{x} \tag{1}$$

where $\boldsymbol{M} \in \mathbb{R}^{\ell n\times \ell n}$ is a block diagonal matrix with the bth diagonal block being the downsampling matrix of band b (as defined in (21)) and $\boldsymbol{B} \in \mathbb{R}^{\ell n\times \ell n}$ is also a block diagonal matrix with the bth diagonal block being a block-circulant-circulant-block (BCCB) matrix for spatially invariant blurring of band b [a 2-D cyclic convolution associated with the point spread function (PSF) of band b, as defined in (20)] [32], [33].

Recovering $\boldsymbol{x}$ from $\boldsymbol{y}$ is obviously an ill-posed inverse problem. To mitigate such ill-posedness, we notice that bands in a multispectral image are often highly correlated, enabling a low-rank representation to reduce the number of variables for representing the target image. For the 12-bands Sentinel-2 data set, we found that more than 99% of the signal energy is retained in the $p = 5$ principal components obtained by a correlation-based eigendecomposition [47]. Therefore, we can write $[\boldsymbol{x}_1, \ldots, \boldsymbol{x}_\ell]^T = \boldsymbol{U}\boldsymbol{Z}$, or equivalently

$$\boldsymbol{x} = (\boldsymbol{U} \otimes \boldsymbol{I}_n)\mathrm{vec}(\boldsymbol{Z}^T) \tag{2}$$

for some full column rank matrix $\boldsymbol{U} \in \mathbb{R}^{\ell\times p}$, which can be learned (from the Sentinel-2 image) ahead of time, as detailed in Section III-B. We term the rows of $\boldsymbol{Z}$ as eigenimages.

From (1) and (2), we focus on the estimation of $\boldsymbol{Z} \in \mathbb{R}^{p\times n}$ based on the following criterion:

$$\min_{\boldsymbol{Z}} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{B}\boldsymbol{x}\|_2^2 + \lambda\phi(\boldsymbol{Z}) \equiv \min_{z} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{B}(\boldsymbol{U} \otimes \boldsymbol{I}_n)z\|_2^2 + \lambda\phi(z) \tag{3}$$

where $z \triangleq \mathrm{vec}(\boldsymbol{Z}^T)$, $\lambda > 0$ is a regularization parameter, and ϕ is a regularizer to further mitigate the ill-posedness [defined in (4)].

[1] A low-dimensional representation of the target image.

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Inspired by the self-similarity prior early rooted in the computer vision area [48], we will explicitly define a new self-similarity concept and accordingly propose a new regularization scheme in Section III. Such a regularization scheme will be incorporated in (3) under the convex optimization framework, which can then be solved without relying on plugging in other self-similarity-guided solvers. A direct advantage is the convergence guarantee of our Sentinel-2 super-resolution algorithm.

III. Convex Regularizer and Convex Solver

In this article, we design ϕ based on the self-similarity prior. The general form is defined as follows:

$$\phi(\boldsymbol{z}) \triangleq \frac{1}{2}\sum_{b=1}^{p}\sum_{(i,j)\in\mathcal{K}} \alpha_{i,j}\,\|\boldsymbol{P}_i\boldsymbol{z}_b - \boldsymbol{P}_j\boldsymbol{z}_b\|_2^2 \tag{4}$$

where $\boldsymbol{z}_b \in \mathbb{R}^n$ is the bth band of the eigenimage $\boldsymbol{Z}$, i.e., $\boldsymbol{z} \equiv \text{vec}(\boldsymbol{Z}^T) = [\boldsymbol{z}_1^T, \ldots, \boldsymbol{z}_p^T]^T$; $\boldsymbol{P}_i \in \mathbb{R}^{q^2\times n}$ serves as an operator that shifts the ith patch from an n-pixel image, where q^2 is the patch size; the self-similarity graph $\mathcal{K}$, collecting pairs of indexes of similar patches, can be learned from the 10-m bands of the observed Sentinel-2 image $\boldsymbol{Y}$, as detailed in Section III-A; $\alpha_{i,j} \geq 0$ measures the similarity between the ith and jth patches in the 10-m bands of $\boldsymbol{Y}$, as detailed in Section III-A. We conclude therefore that the regularizer ϕ promotes solutions $\boldsymbol{z}_b$ the degree of similarity between the patches $\boldsymbol{P}_i\boldsymbol{z}_b$ and $\boldsymbol{P}_j\boldsymbol{z}_b$ of which is set by the value of $\alpha_{i,j}$; higher values of $\alpha_{i,j}$ correspond to higher degrees of similarity.

The design of the self-similarity regularization is highly flexible under the general framework specified by (4), for which the self-similarity graph $\mathcal{K}$ and the similarity measure $\alpha_{i,j}$ can be customized for other imaging inverse problems. For the Sentinel-2 super-resolution task, the exact definitions of $\mathcal{K}$ and $\alpha_{i,j}$ are given in Section III-A.

The metric of the distance between two patches is also flexible, as long as it is a convex function; this is formally stated in the following fact.

Fact 1: If the distance metric $\text{dist}(\boldsymbol{p}_1, \boldsymbol{p}_2)$ is convex in $(\boldsymbol{p}_1, \boldsymbol{p}_2)$ and $\alpha_{i,j} \geq 0$, then the self-similarity regularizer

$$\phi(\boldsymbol{z}) \triangleq \frac{1}{2}\sum_{b=1}^{p}\sum_{(i,j)\in\mathcal{K}} \alpha_{i,j}\,\text{dist}(\boldsymbol{P}_i\boldsymbol{z}_b, \boldsymbol{P}_j\boldsymbol{z}_b),$$

which is a generalized version of (4), is still convex. □

Just like there are several variations of TV regularization, we leave sufficient space for exploring other variations of the important self-similarity regularization while bringing it into the convex framework. In other words, the metric measuring the distance between two patches might be replaced by any convex function if necessary. The proof of Fact 1 straightforwardly follows from the propositions that any composition of a convex function [e.g., $\text{dist}(\boldsymbol{p}_1, \boldsymbol{p}_2) = \|\boldsymbol{p}_1 - \boldsymbol{p}_2\|_2^2$ in (4)] with an affine function [e.g., $\boldsymbol{P}_i\boldsymbol{z}_b - \boldsymbol{P}_j\boldsymbol{z}_b$ in (4)] is a convexity-preserving operator [45] and nonnegative weighted sum of convex functions also preserves convexity [45]. Note that our convergence analysis is generally applicable to tackle such flexibility; it does not pose any assumption on the graph and similarity measure, as stated in Theorem 1.

It is interesting to note that (4) incorporates the self-similarity prior by casting it into a form of Tikhonov regularization [23], so one actually can derive the closed-form solution of (3) directly. However, such a closed-form solution requires computing the inverse of a huge matrix, which is not computationally feasible, so we need to design a computationally efficient solver for practical applicability.

Specifically, considering that (3) is a large-scale convex optimization problem, we solve it by the powerful convex optimization tool—SALSA [42], which is an instance of ADMM [45]. In order to exploit the decomposability of SALSA, we equivalently reformulate (3) as

$$\begin{aligned}\min_{\boldsymbol{z},\boldsymbol{v},\{\boldsymbol{V}_{i,j}\}}\ & \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{B}\boldsymbol{v}\|_2^2 + \frac{\lambda}{2}\sum_{(i,j)\in\mathcal{K}}\alpha_{i,j}\|(\boldsymbol{P}_i - \boldsymbol{P}_j)\boldsymbol{V}_{i,j}\|_F^2\\ \text{s.t.}\quad & \boldsymbol{v} = (\boldsymbol{U}\otimes\boldsymbol{I}_n)\boldsymbol{z}\\ & \boldsymbol{V}_{i,j} = \boldsymbol{Z}^T\quad \forall (i,j)\in\mathcal{K}\end{aligned} \tag{5}$$

the augmented Lagrangian of which is

$$\begin{aligned}&\mathcal{L}(\boldsymbol{v}, \{\boldsymbol{V}_{i,j}\}, \boldsymbol{z}, \boldsymbol{d}, \{\boldsymbol{D}_{i,j}\})\\ &= \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{B}\boldsymbol{v}\|_2^2 + \frac{\mu}{2}\|(\boldsymbol{U}\otimes\boldsymbol{I}_n)\boldsymbol{z} - \boldsymbol{v} - \boldsymbol{d}\|_2^2\\ &\quad + \sum_{(i,j)\in\mathcal{K}}\left(\frac{\lambda\alpha_{i,j}}{2}\|(\boldsymbol{P}_i - \boldsymbol{P}_j)\boldsymbol{V}_{i,j}\|_F^2 + \frac{\mu}{2}\|\boldsymbol{Z}^T - \boldsymbol{V}_{i,j} - \boldsymbol{D}_{i,j}\|_F^2\right)\end{aligned}$$

where $\boldsymbol{d}$ and $\{\boldsymbol{D}_{i,j}\}$ are scaled dual variables and $\mu > 0$ is called penalty parameter. Then, the SALSA algorithm for solving (5) is summarized in Algorithm 1, which is referred to as SSSS, the convergence guarantee of which is stated in Theorem 1 and proved in Appendix A.

Algorithm 1 SSSS Algorithm for Solving (5)

1: **Given** $(\boldsymbol{M}, \boldsymbol{B}, \boldsymbol{U}, \boldsymbol{y})$, $(\mathcal{K}, \alpha_{i,j})$, $\lambda > 0$ and $\mu > 0$.
2: Initialize $\boldsymbol{z}^0 := \boldsymbol{0}_{pn}$, $\boldsymbol{d}^0 := \boldsymbol{0}_{\ell n}$, $\boldsymbol{D}_{i,j}^0 := \boldsymbol{0}_{n\times p}$ (or warm start). Set $k := 0$.
3: **repeat**
4: Update $\left(\boldsymbol{v}^{k+1}, \left\{\boldsymbol{V}_{i,j}^{k+1}\right\}\right) \in \arg\min_{\boldsymbol{v},\{\boldsymbol{V}_{i,j}\}}\ \mathcal{L}\left(\boldsymbol{v}, \{\boldsymbol{V}_{i,j}\}, \boldsymbol{z}^k, \boldsymbol{d}^k, \{\boldsymbol{D}_{i,j}^k\}\right)$;
5: Update $\boldsymbol{z}^{k+1} \in \arg\min_{\boldsymbol{z}}\ \mathcal{L}\left(\boldsymbol{v}^{k+1}, \{\boldsymbol{V}_{i,j}^{k+1}\}, \boldsymbol{z}, \boldsymbol{d}^k, \{\boldsymbol{D}_{i,j}^k\}\right)$;
6: Update $\boldsymbol{d}^{k+1} := \boldsymbol{d}^k + \boldsymbol{v}^{k+1} - (\boldsymbol{U}\otimes\boldsymbol{I}_n)\boldsymbol{z}^{k+1}$ and $\boldsymbol{D}_{i,j}^{k+1} := \boldsymbol{D}_{i,j}^k + \boldsymbol{V}_{i,j}^{k+1} - (\boldsymbol{Z}^{k+1})^T$;
7: $k := k + 1$;
8: **until** the predefined stopping criterion is met.
9: **Output** the super-resolved Sentinel-2 image $\widehat{\boldsymbol{x}} \equiv \text{vec}(\widehat{\boldsymbol{X}}^T) := (\boldsymbol{U}\otimes\boldsymbol{I}_n)\boldsymbol{z}^k$ (cf. (2)).

Theorem 1: Consider any graph $\mathcal{K} \subseteq \{(i,j) \mid i, j \in \mathcal{I}_N, i \neq j\}$ and $\alpha_{i,j} \geq 0,\ \forall (i,j) \in \mathcal{K}$, where N is the number of patches. The sequence $\{\boldsymbol{z}^k\}$ generated by Algorithm 1 converges to an optimal solution of the self-similarity regularized problem (5). □

Next, we will derive solutions for the optimization problems involved in lines 4 and 5 of Algorithm 1 and detail the basis

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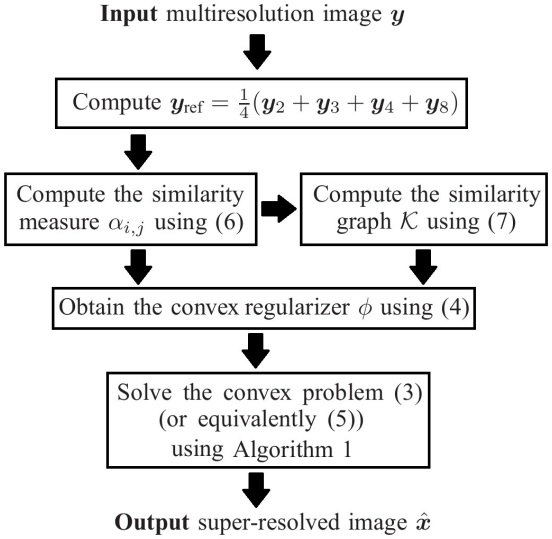


Fig. 1. Flowchart of similarity graph learning and the proposed method.

learning (for obtaining $\boldsymbol{U}$) and graph learning (for obtaining $\alpha_{i,j}$ and $\mathcal{K}$).

A. Graph Learning

In this section, we propose a natural way to learn the self-similarity graph $\mathcal{K}$ from some bands of highest resolution (i.e., B2–B4 and B8) in the observed Sentinel-2 image. To this end, we have to measure the similarity $\alpha_{i,j}$ between the ith patch and jth patch in the reference image defined by the four finest resolution bands; without additional information, we simply take the average of the four finest resolution bands as the reference image, i.e., $\boldsymbol{y}_{\text{ref}} = (1/4)(\boldsymbol{y}_2 + \boldsymbol{y}_3 + \boldsymbol{y}_4 + \boldsymbol{y}_8)$. As the self-similarity graph $\mathcal{K}$ (defined next) will be learned based on $\boldsymbol{y}_{\text{ref}}$, its scale is also defined at the finest resolution (i.e., 10 m). Basically, the similarity measure $\alpha_{i,j}$ should be nonnegative and be a decreasing function of the distance between the ith patch (i.e., $\boldsymbol{P}_i \boldsymbol{y}_{\text{ref}}$) and the jth patch (i.e., $\boldsymbol{P}_j \boldsymbol{y}_{\text{ref}}$) in the reference image. Here, we simply define it as

$$\alpha_{i,j} \triangleq \|\boldsymbol{P}_i \boldsymbol{y}_{\text{ref}} - \boldsymbol{P}_j \boldsymbol{y}_{\text{ref}}\|_2^{-1}. \tag{6}$$

Then, we learn the self-similarity graph as follows:

$$\mathcal{K} \triangleq \left\{(i,j) \mid i \in \mathcal{I}_N,\ j \in \arg \max_{j' \in \mathcal{I}_N,\ j' \neq i} \alpha_{i,j'}\right\} \tag{7}$$

where N is the number of patches. In other words, for each patch i, we find its most similar (but distinct) patch j, and then add an edge (i, j) to the graph $\mathcal{K}$. As will be seen, this simple graph learning procedure is quite effective in building the self-similarity pattern for regularizing the Sentinel-2 super-resolution problem. A flowchart of the proposed method (including the learning of $\mathcal{K}$) is given in Fig. 1.

B. Basis Learning

In this section, we learn the basis $\boldsymbol{U} \in \mathbb{R}^{\ell \times p}$ for representing the target image $[\boldsymbol{x}_1, \ldots, \boldsymbol{x}_\ell]^T = \boldsymbol{U}\boldsymbol{Z} \in \mathbb{R}^{\ell \times n}$ in a p-dimensional subspace. Combining (1) and (2), it is natural to select the basis $\boldsymbol{U}$ as the argument that minimizes the fitting error $\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{B}(\boldsymbol{U} \otimes \boldsymbol{I}_n)\boldsymbol{z}\|_2^2$, where $\boldsymbol{z}$ is an unknown variable and thus replaced by $\widetilde{\boldsymbol{z}} = \boldsymbol{z}^{30}$ (i.e., the intermediate variable $\boldsymbol{z}^k$ obtained by SSSS at the 30th iteration), which already yields meaningful images according to our empirical experience ($\boldsymbol{U}$ is simply set by principal components analysis [45] in the first 30 iterations). To be precise, the basis is learned by solving the following optimization problem:

$$\boldsymbol{U} \in \arg \min_{\boldsymbol{U}' \in \mathbb{R}^{\ell \times p}} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{B}(\boldsymbol{U}' \otimes \boldsymbol{I}_n)\widetilde{\boldsymbol{z}}\|_2^2. \tag{8}$$

Let $\boldsymbol{U} \equiv [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_\ell]^T$. Then, (8) can be decoupled as ℓ subproblems, that is

$$\min_{\boldsymbol{u}_b \in \mathbb{R}^p} \frac{1}{2}\|\boldsymbol{y}_b - \boldsymbol{M}_b\boldsymbol{B}_b\widetilde{\boldsymbol{Z}}^T\boldsymbol{u}_b\|_2^2, \quad b = 1, \ldots, \ell \tag{9}$$

where $\boldsymbol{M}_b$ (resp., $\boldsymbol{B}_b$) is the bth diagonal block of $\boldsymbol{M}$ (resp., $\boldsymbol{B}$) and $\widetilde{\boldsymbol{Z}}$ satisfies $\widetilde{\boldsymbol{z}} \triangleq \text{vec}(\widetilde{\boldsymbol{Z}}^T)$. The closed-form solution of (9) is then given by

$$\boldsymbol{u}_b^\star := \left(\widetilde{\boldsymbol{Z}}\boldsymbol{B}_b^T\boldsymbol{M}_b^T\boldsymbol{M}_b\boldsymbol{B}_b\widetilde{\boldsymbol{Z}}^T\right)^{-1}\left(\widetilde{\boldsymbol{Z}}\boldsymbol{B}_b^T\boldsymbol{M}_b^T\boldsymbol{y}_b\right).$$

C. Solution for Line 4 in Algorithm 1

Note that the variables involved in line 4, i.e., $\boldsymbol{v}$ and $\boldsymbol{V}_{i,j}$, can be decoupled as

$$\begin{aligned}\boldsymbol{v}^{k+1} \in \arg\min_{\boldsymbol{v}} &\frac{1}{2}\|\boldsymbol{y} - \boldsymbol{M}\boldsymbol{B}\boldsymbol{v}\|_2^2 \\ &+ \frac{\mu}{2}\|(\boldsymbol{U} \otimes \boldsymbol{I}_n)\boldsymbol{z}^k - \boldsymbol{v} - \boldsymbol{d}^k\|_2^2 \end{aligned} \tag{10}$$

$$\begin{aligned}\boldsymbol{V}_{i,j}^{k+1} \in \arg\min_{\boldsymbol{V}_{i,j}} &\frac{\lambda\alpha_{i,j}}{2}\|(\boldsymbol{P}_i - \boldsymbol{P}_j)\boldsymbol{V}_{i,j}\|_F^2 \\ &+ \frac{\mu}{2}\left\|(\boldsymbol{Z}^k)^T - \boldsymbol{V}_{i,j} - \boldsymbol{D}_{i,j}^k\right\|_F^2. \end{aligned} \tag{11}$$

Next, we solve (10) and (11) in Sections III-C1 and III-C2, respectively.

1) Solution for (10)*:* Let $\boldsymbol{v} \triangleq [\boldsymbol{v}_1^T, \ldots, \boldsymbol{v}_\ell^T]^T$. The optimization (10) can be further decoupled for each band $b = 1, \ldots, \ell$

$$\boldsymbol{v}_b^{k+1} \in \arg\min_{\boldsymbol{v}_b} \frac{1}{2}\|\boldsymbol{y}_b - \boldsymbol{M}_b\boldsymbol{B}_b\boldsymbol{v}_b\|_2^2 + \frac{\mu}{2}\|\boldsymbol{v}_b - \bar{\boldsymbol{v}}_b\|_2^2 \tag{12}$$

where $\bar{\boldsymbol{v}}_b \triangleq [(\boldsymbol{U} \otimes \boldsymbol{I}_n)\boldsymbol{z}^k - \boldsymbol{d}^k]_{1+(b-1)n:bn}$, and $\boldsymbol{M}_b$ and $\boldsymbol{B}_b$ are the bth diagonal block of $\boldsymbol{M}$ and $\boldsymbol{B}$, respectively. A closed-form solution of (12) is given by

$$\boldsymbol{v}_b^{k+1} = \left(\boldsymbol{B}_b^T\boldsymbol{M}_b^T\boldsymbol{M}_b\boldsymbol{B}_b + \mu\boldsymbol{I}_n\right)^{-1}\left(\boldsymbol{B}_b^T\boldsymbol{M}_b^T\boldsymbol{y}_b + \mu\bar{\boldsymbol{v}}_b\right) \tag{13}$$

which is not computationally efficient (due to the inversion of a huge $n \times n$ matrix) and has to be further refined.

The aim is to avoid computing the inversion of a huge matrix. A technique developed in [49] can be adopted for such purpose, and it is concisely recalled here for a self-contained purpose. Assume that the target image $\boldsymbol{X}$ contains $n_1' \times n_2'$ pixels (so $n = n_1'n_2'$). Using the Woodbury matrix identity [45] and Proposition 1 (see Appendix B), the matrix inversion in (13) can be refined as

$$\begin{aligned}&\left(\boldsymbol{B}_b^T\boldsymbol{M}_b^T\boldsymbol{M}_b\boldsymbol{B}_b + \mu\boldsymbol{I}_n\right)^{-1} \\ &\quad= \frac{1}{\mu}\boldsymbol{I}_n - \frac{1}{\mu^2}\boldsymbol{F}^H\boldsymbol{\Sigma}^H\boldsymbol{D}\left(r_i^2\boldsymbol{I}_{n_i} + \frac{1}{\mu}\boldsymbol{D}^H|\boldsymbol{\Sigma}^2|\boldsymbol{D}\right)^{-1}\boldsymbol{D}^H\boldsymbol{\Sigma}\boldsymbol{F}\end{aligned} \tag{14}$$

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as proved in Appendix B, where $\boldsymbol{F} \triangleq \boldsymbol{F}_{n_2'} \otimes \boldsymbol{F}_{n_1'} \in \mathbb{C}^{n\times n}$, $\boldsymbol{D} \triangleq (\mathbf{1}_{r_i} \otimes \boldsymbol{I}_{n_2'/r_i}) \otimes (\mathbf{1}_{r_i} \otimes \boldsymbol{I}_{n_1'/r_i}) \in \mathbb{R}^{n\times n_i}$, $\boldsymbol{\Sigma}$ is a diagonal matrix with its diagonal vector $\mathrm{diag}(\boldsymbol{\Sigma})$ given by the vectorized version of the 2-D DFT of the $n_1' \times n_2'$ convolution blurring kernel [see (20)], and r_i and n_i are the upsampling factor of the band b and the number of pixels in the band b, resp. The 2-D DFT in (14) (i.e., $\boldsymbol{F}$) can be very efficiently implemented by fast Fourier transform, if the image is not too large; otherwise, one would simply divide it into subimages and separately super-resolve them.

A remarkable property is that the matrix inversion involved in (13) can now be efficiently computed using (14), because $(r_i^2 \boldsymbol{I}_{n_i} + (1/\mu)\boldsymbol{D}^H|\boldsymbol{\Sigma}^2|\boldsymbol{D})$ is diagonal (making its inversion easily computable). Such diagonality is not immediately clear from [49], and thus, a short proof is given in Appendix C for completeness.

2) Solution for (11)*:* Letting $\boldsymbol{v}_{i,j}^{(c)}$ be the cth column of $\boldsymbol{V}_{i,j}^{k+1}$, (11) can be decoupled with respect to columns

$$\min_{\boldsymbol{v}_{i,j}^{(c)}} \frac{\lambda\alpha_{i,j}}{2}\left\|(\boldsymbol{P}_i - \boldsymbol{P}_j)\boldsymbol{v}_{i,j}^{(c)}\right\|_2^2 + \frac{\mu}{2}\left\|\boldsymbol{v}_{i,j}^{(c)} - \boldsymbol{\delta}_{i,j}^{(c)}\right\|_2^2 \tag{15}$$

where $\boldsymbol{\delta}_{i,j}^{(c)}$ denotes the cth column of $(\boldsymbol{Z}^k)^T - \boldsymbol{D}_{i,j}^k$. Note that since $(\boldsymbol{P}_i - \boldsymbol{P}_j) \in \mathbb{R}^{q^2\times n}$ (the patch size is typically set as $q \times q \equiv 6 \times 6$ in this article), directly solving (15) requires computing the inverse of a large $n \times n$ matrix—not computationally affordable.

To solve (15) efficiently, we notice that $(\boldsymbol{P}_i - \boldsymbol{P}_j) \in \mathbb{R}^{q^2\times n}$ is actually a column-sparse matrix. Let $\mathcal{I} \subseteq \mathcal{I}_n$ be the set collecting indices of nonzero columns of the matrix $(\boldsymbol{P}_i - \boldsymbol{P}_j)$. We have $|\mathcal{I}| \leq 2q^2$ because each row of $\boldsymbol{P}_i$ (and $\boldsymbol{P}_j$) has exactly one element being 1 with other elements being 0. Then, we can reformulate (15) as (16), as shown at the bottom of this page, where $[\boldsymbol{v}]_{\mathcal{S}}$ is a subvector of $\boldsymbol{v}$ formed by collecting elements with indices specified by the index set $\mathcal{S}$, $\bar{\mathcal{I}} \triangleq \mathcal{I}_n \setminus \mathcal{I}$, and $[\boldsymbol{P}_i - \boldsymbol{P}_j]_{\mathcal{I}} \in \mathbb{R}^{q^2\times|\mathcal{I}|}$ is a trimmed version of $(\boldsymbol{P}_i - \boldsymbol{P}_j)$ by removing zero columns. A computationally efficient closed-form solution is hence obtained as follows:

$$\left[\boldsymbol{v}_{i,j}^{(c)}\right]_{\mathcal{I}}^{\star} = \left[[\boldsymbol{P}_i - \boldsymbol{P}_j]_{\mathcal{I}}^T[\boldsymbol{P}_i - \boldsymbol{P}_j]_{\mathcal{I}} + \frac{\mu\boldsymbol{I}_{|\mathcal{I}|}}{\lambda\alpha_{i,j}}\right]^{-1}\left(\frac{\mu\left[\boldsymbol{\delta}_{i,j}^{(c)}\right]_{\mathcal{I}}}{\lambda\alpha_{i,j}}\right)$$

$$\left[\boldsymbol{v}_{i,j}^{(c)}\right]_{\bar{\mathcal{I}}}^{\star} = \left[\boldsymbol{\delta}_{i,j}^{(c)}\right]_{\bar{\mathcal{I}}}$$

for which we only need to compute the inverse of a small $|\mathcal{I}| \times |\mathcal{I}|$ matrix with $|\mathcal{I}| \leq 2q^2$.

D. Solution for Line 5 in Algorithm 1

The problem can be simplified as

$$\boldsymbol{z}^{k+1} \in \arg\min_{\boldsymbol{z}} \|(\boldsymbol{U} \otimes \boldsymbol{I}_n)\boldsymbol{z} - \boldsymbol{v}^{k+1} - \boldsymbol{d}^k\|_2^2 + \sum_{(i,j)\in\mathcal{K}} \left\|\boldsymbol{Z}^T - \boldsymbol{V}_{i,j}^{k+1} - \boldsymbol{D}_{i,j}^k\right\|_F^2. \tag{17}$$

Define $\boldsymbol{\Psi}_0 \triangleq (\boldsymbol{U} \otimes \boldsymbol{I}_n)^T(\boldsymbol{U} \otimes \boldsymbol{I}_n) + |\mathcal{K}|\boldsymbol{I}_{pn}$, $\boldsymbol{\Psi}_1 \triangleq \mathrm{vec}_{n\times\ell}^{-1}(\boldsymbol{v}^{k+1} + \boldsymbol{d}^k)\,\boldsymbol{U}$, and $\boldsymbol{\Psi}_2 \triangleq \sum_{(i,j)\in\mathcal{K}}(\boldsymbol{V}_{i,j}^{k+1} + \boldsymbol{D}_{i,j}^k)$. Then, the closed-form solution of (17) can be derived as

$$\boldsymbol{z}^{k+1} = \boldsymbol{\Psi}_0^{-1}[(\boldsymbol{U} \otimes \boldsymbol{I}_n)^T(\boldsymbol{v}^{k+1} + \boldsymbol{d}^k) + \mathrm{vec}(\boldsymbol{\Psi}_2)] \tag{18}$$

$$= \mathrm{vec}((\boldsymbol{\Psi}_1 + \boldsymbol{\Psi}_2)(\boldsymbol{U}^T\boldsymbol{U} + |\mathcal{K}|\boldsymbol{I}_p)^{-1}) \tag{19}$$

where (19) is much more computationally efficient than (18) as the involved matrix inversion associates with a much smaller $p \times p$ matrix.

IV. Experimental Results and Discussion

In this section, we super-resolve real Sentinel-2 images using the proposed SSSS algorithm, which is for visual quality assessment only, as the ground truth is not available. To compare with peer methods, we simulate the ground-truth images (i.e., simulated Sentinel-2 images with each band having 10-m GSD) using real hyperspectral images, and then, the ground-truth images are blurred and subsampled according to (1) to generate the simulated Sentinel-2 images, which are then processed by Sentinel-2 super-resolution methods for quantitative comparisons. For both real and simulated images, the hyperparameters in Algorithm 1 (with 100 iterations) are unified as $(\lambda, \mu) = (0.1, 0.1)$, which are obtained by grid search and may not be optimal. Among the peer methods, MUSA [33] and S2Sharp [24] also involve hyperparameters. For MUSA (with 130 iterations), its augmented Lagrange parameter and regularization parameter are obtained by grid search and, respectively, set as 0.6 and 0.005. For S2Sharp (with ten iterations), its model order (i.e., rank) and the weight vector in its cost function (i.e., [24, eq. (2)]) are set as default values (in the source codes released by its authors), i.e., 8 and $(1e-3)$[0.19, 0.07, 1.31, 3.79, 9.10, 5.23, 0.51, 6.62], respectively. We remark that the high dimensionality of hyperparameter space of S2Sharp disables effective use of naive grid search; though we also tried the more sophisticated generalized cross validation (GCV) [24], it did not yield better hyperparameters than the default ones. The details will be presented in the subsequent sections.

A. Real Data Experiment

The real Sentinel-2 data contain four high-resolution bands of 10-m GSD (i.e., B2–B4 and B8), six medium-resolution bands of 20-m GSD (i.e., B5–B7, B8a, B11, and B12), and two low-resolution bands of 60-m GSD (i.e., B1 and B9); note that band B10 is not considered here as it records only the cirrus information for atmospheric correction [46]. We investigate four commonly studied geographical areas on the Earth surface, including crop area, coastal area, mountainous area, and urban area. The crop data set was acquired on March 11, 2017 (Tile T10SEH), while the other three data

$$\min_{\left[\boldsymbol{v}_{i,j}^{(c)}\right]_{\mathcal{I}},\left[\boldsymbol{v}_{i,j}^{(c)}\right]_{\bar{\mathcal{I}}}} \left\{\frac{\lambda\alpha_{i,j}}{2}\left\|[\boldsymbol{P}_i - \boldsymbol{P}_j]_{\mathcal{I}}\left[\boldsymbol{v}_{i,j}^{(c)}\right]_{\mathcal{I}}\right\|_2^2 + \frac{\mu}{2}\left\|\left[\boldsymbol{v}_{i,j}^{(c)}\right]_{\mathcal{I}} - \left[\boldsymbol{\delta}_{i,j}^{(c)}\right]_{\mathcal{I}}\right\|_2^2 + \frac{\mu}{2}\left\|\left[\boldsymbol{v}_{i,j}^{(c)}\right]_{\bar{\mathcal{I}}} - \left[\boldsymbol{\delta}_{i,j}^{(c)}\right]_{\bar{\mathcal{I}}}\right\|_2^2\right\} \tag{16}$$

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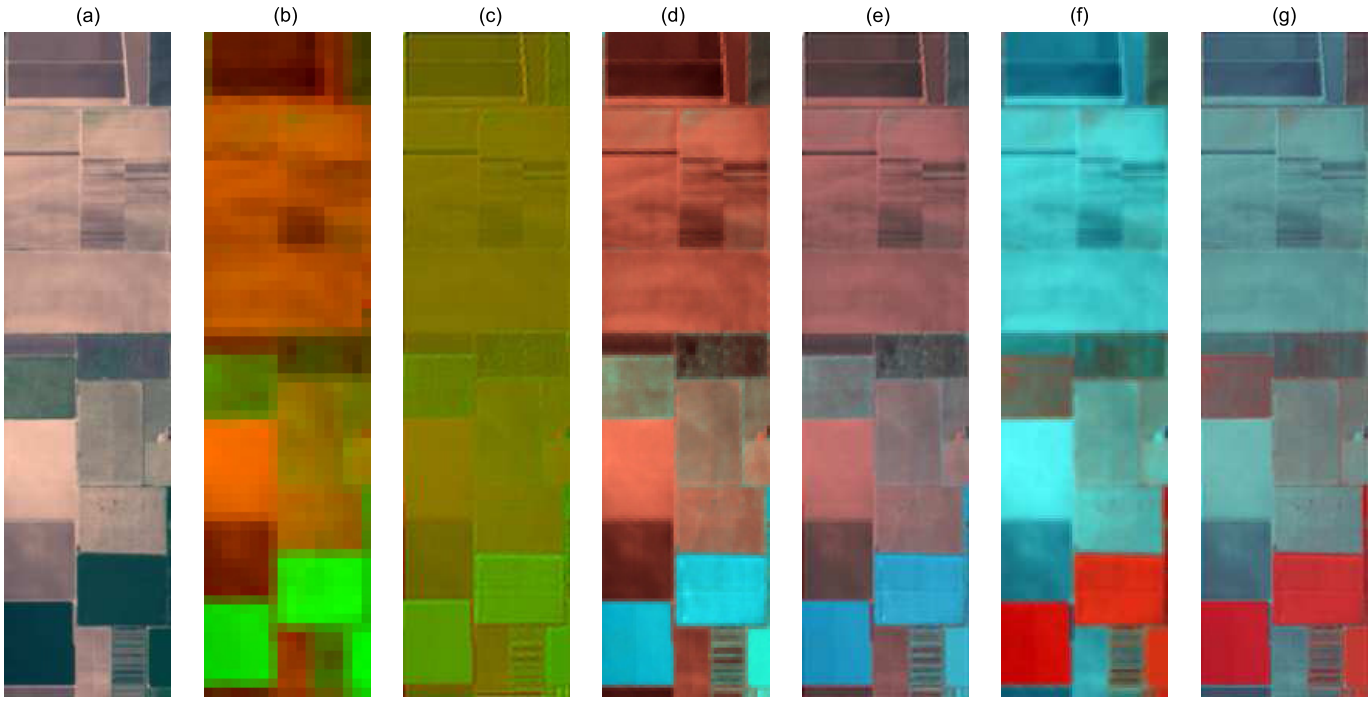


Fig. 2. Real Sentinel-2 data set for crop area. (a) True color composition (10 m). (b) False-color composition of bands 1 and 9 (60 m). (d) False-color composition of bands 5–7 (20 m). (f) False-color composition of bands 8a, 11, and 12 (20 m). (c), (e), and (g) Super-resolved versions (obtained by SSSS) corresponding to (b), (d), and (f), respectively.

sets were acquired on September 11, 2017 (Tile T11SMS). The 10-m bands of these four data sets all contain 432×108 pixels, and their true color compositions are shown in Figs. 2(a), 3(a), 4(a), and 5(a).

These data sets are processed by the proposed SSSS algorithm for super-resolving the 60-/20-m bands to achieve 10-m spatial resolution. The false-color compositions of the 60-m bands of these four data sets are shown in Figs. 2(b), 3(b), 4(b), and 5(b); their super-resolved versions are then shown in Figs. 2(c), 3(c), 4(c), and 5(c). The false-color compositions of the first three 20-m bands of these data sets are shown in Figs. 2(d), 3(d), 4(d), and 5(d); their super-resolved versions are then shown in Figs. 2(e), 3(e), 4(e), and 5(e). Similarly, the false-color compositions of the last three 20 m bands of these data sets are shown in Figs. 2(f), 3(f), 4(f), and 5(f), respectively; their super-resolved versions are then shown in Figs. 2(g), 3(g), 4(g), and 5(g), respectively.

From Figs. 2–5, one can clearly see that SSSS provides very good qualitative results. For example, the textures of bands 1 and 9 of the mountainous area [see Fig. 4(c)] are successfully reconstructed from the quite blurred input images of 60-m GSD [see Fig. 4(b)]. For another example, the spatial details for the urban area are also well recovered from those low-/medium-resolution bands, as characterized by the high-frequency details of the buildings [see Fig. 5(b) and (c)]. Similar comments apply to the areas containing crops and coastal line (see Figs. 2 and 3). Note that since there is no available ground truth for the real data experiment, we only show the visual results to confirm the applicability of SSSS to real Sentinel-2 data. The purpose of Figs. 2–5 is merely to show that the newly discovered convex regularization scheme [see (4)] does function normally in various land covers. As peer methods, all yield products with good visual quality, showing that their visual results are not a good way to understand the advantages of SSSS, for which one still need resort to quantitative results. For example, a key advantage of SSSS is the capability of preserving the spectral property (see Section IV-C), which is hard to be illustrated visually but can be clearly seen from the quantitative results (see spectral angle mapper (SAM) values in Table II). We will show comprehensive quantitative results in the subsequent sections.

B. Preparation of Simulated Sentinel-2 Data

We generate the simulated data using the standard procedure in the literature [33], following Wald's protocol and ESA's data quality report on Sentinel-2 satellite products [50], [51]. For the self-contained purpose, we concisely recall the synthetic data generation procedure here.

First, we describe how to generate the ground truth, i.e., the 12-band target image $\boldsymbol{x}$ of 10-m GSD, which is unknown to the algorithms under test. It is based on 224-band hyperspectral images collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor [52], which is suitable for our purpose as its spectral range (i.e., 0.4–2.5 μm) covers that of Sentinel-2 images. Four AVIRIS images, corresponding to spatial regions of crop (acquired on August 2013 at 3.2-m GSD), coastal (acquired on November 2011 at 5-m GSD), mountainous (acquired on November 2011 at 5-m GSD), and urban (acquired on May 2006 at 3.5-m GSD) areas, are selected according to the four scenes studied in Section IV-A. The true-color composition of the four AVIRIS

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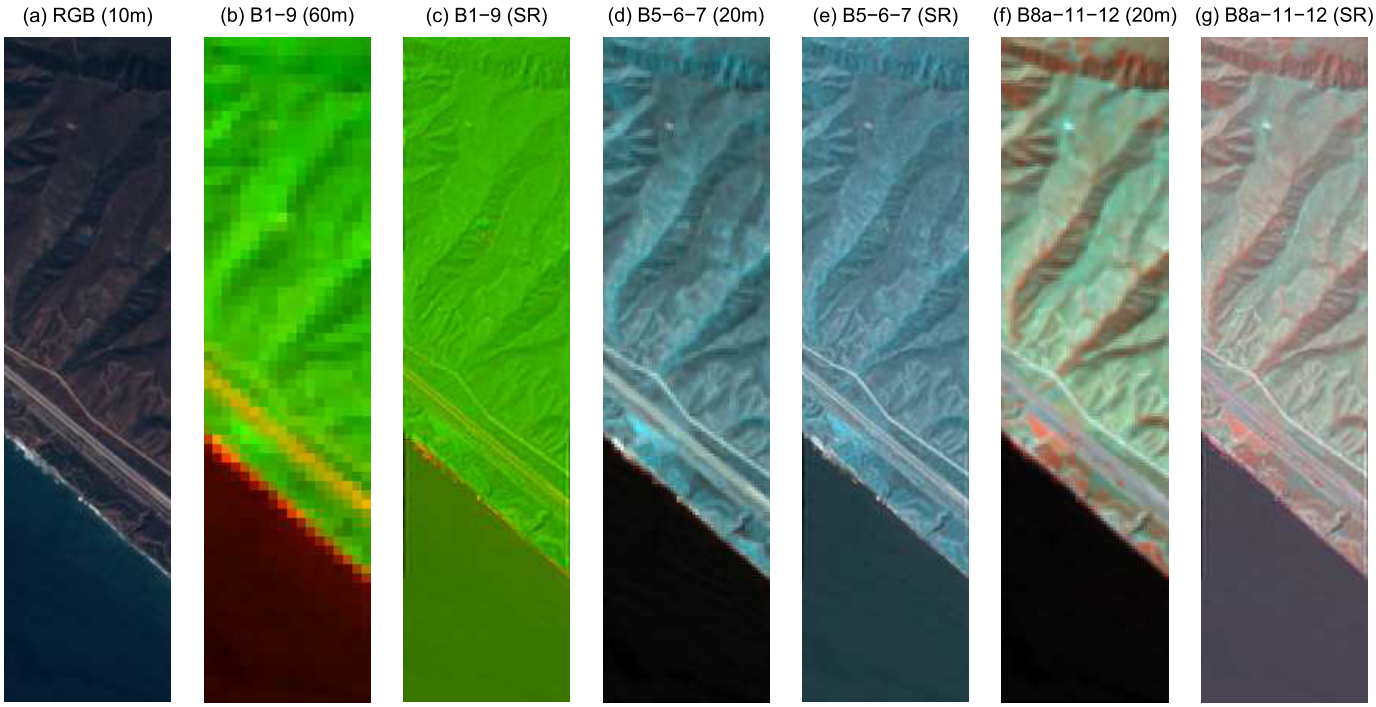


Fig. 3. Real Sentinel-2 data set for coastal area. (a) True color composition (10 m). (b) False-color composition of bands 1 and 9 (60 m). (d) False-color composition of bands 5–7 (20 m). (f) False-color composition of bands 8a, 11, and 12 (20 m). (c), (e), and (g) Super-resolved versions (obtained by SSSS) corresponding to (b), (d), and (f).

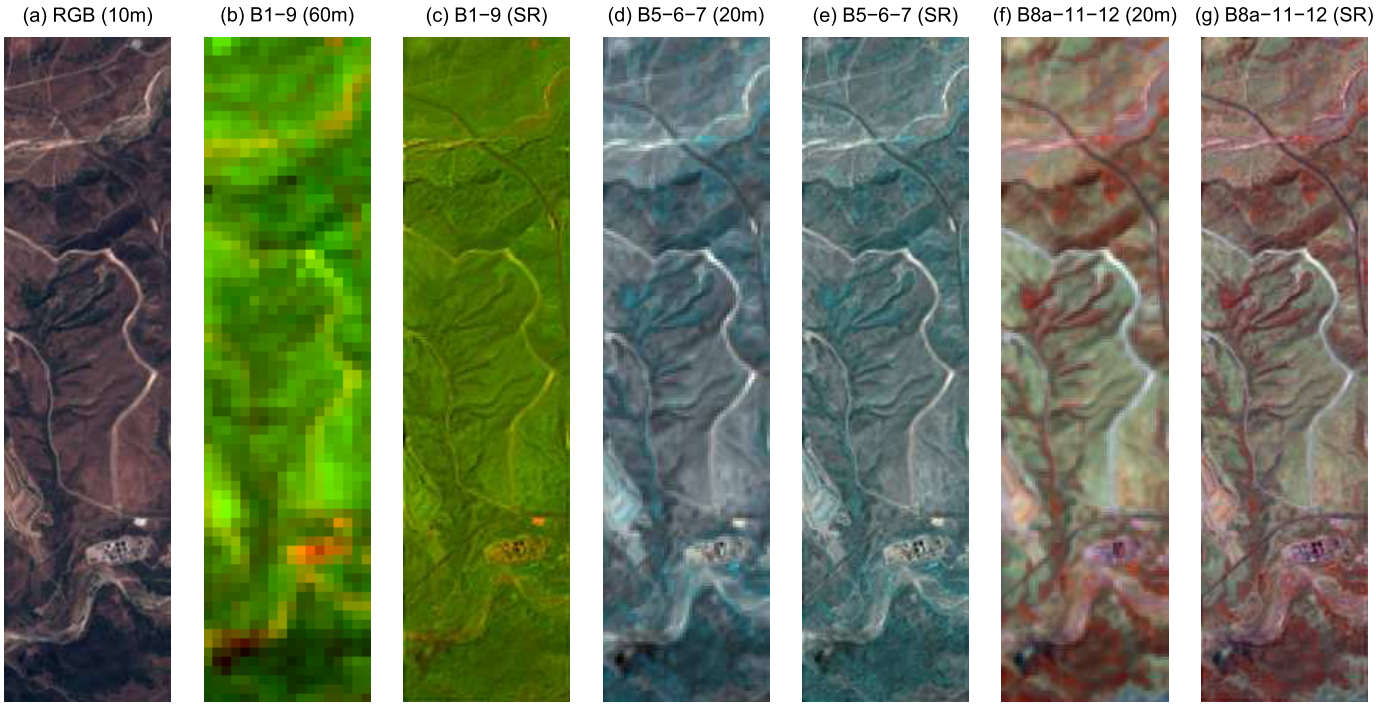


Fig. 4. Real Sentinel-2 data set for mountainous area. (a) True color composition (10 m). (b) False-color composition of bands 1 and 9 (60 m). (d) False-color composition of bands 5–7 (20 m). (f) False-color composition of bands 8a, 11, and 12 (20 m). (c), (e) and (g) Super-resolved versions (obtained by SSSS) corresponding to (b), (d), and (f), respectively.

scenes is shown in Fig. 6 [see Figs. 2(a), 3(a), 4(a), and 5(a)]. The AVIRIS images are then filtered by 15×15 Gaussian kernels (with standard deviation of 1.6 for crop data, 1.2 for coastal and mountainous data, and 1.5 for urban data), followed by spatial downsampling (with a factor of 3 for crop data, 2 for coastal and mountainous data, and 3 for urban data) for approximately reaching hyperspectral images of 10-m GSD, denoted as $\boldsymbol{x}_h$. Then, the spectral properties

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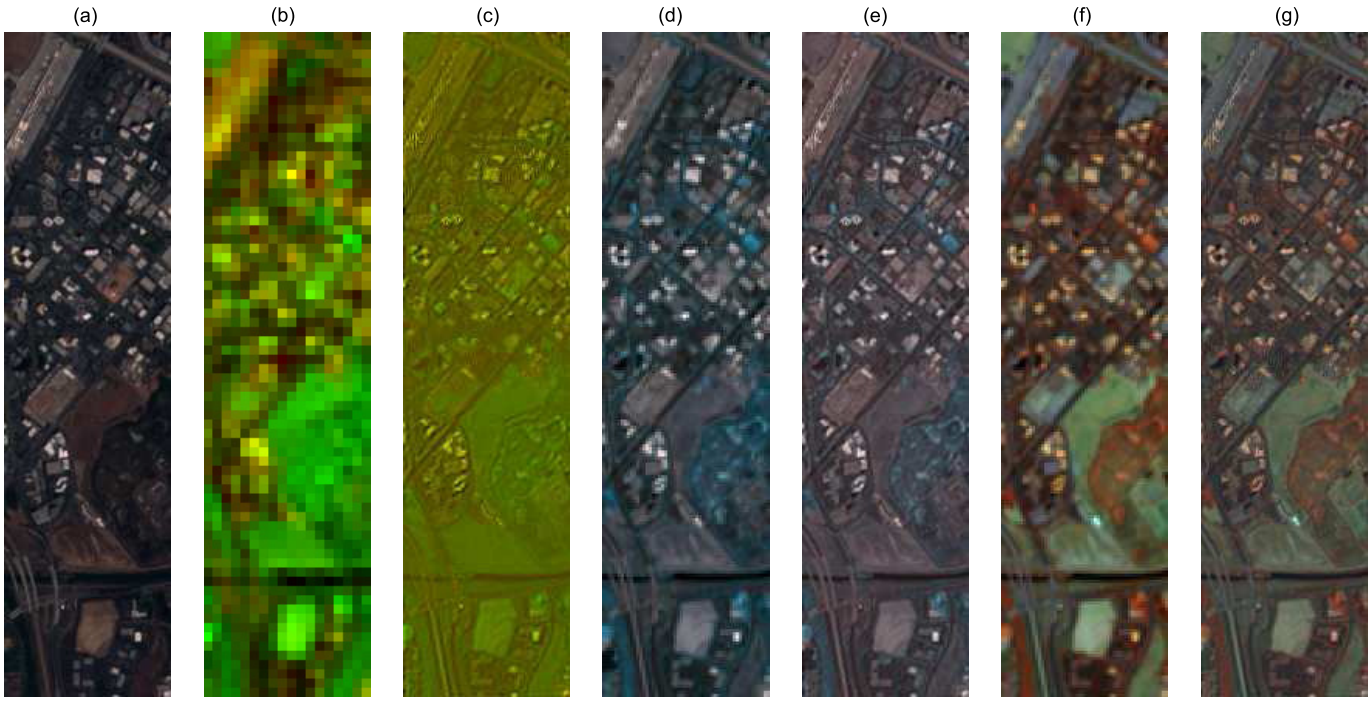


Fig. 5. Real Sentinel-2 data set for urban area. (a) True color composition (10 m). (b) False-color composition of bands 1 and 9 (60 m). (d) False-color composition of bands 5–7 (20 m). (f) False-color composition of bands 8a, 11, and 12 (20 m). (c), (e), and (g) Super-resolved versions (obtained by SSSS) corresponding to (b), (d), and (f), respectively.

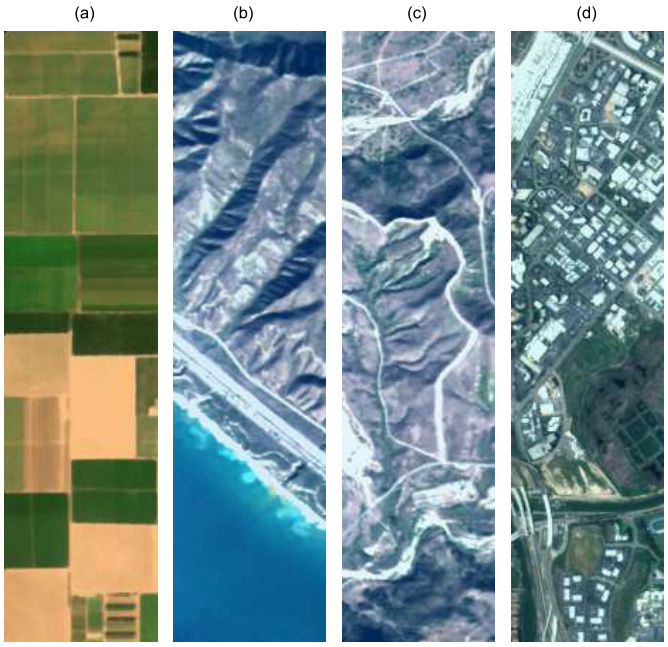


Fig. 6. True-color composition of AVIRIS-simulated Sentinel-2 data set for (a) crop area, (b) coastal area, (c) mountainous area, and (d) urban area.

of Sentinel-2 images are simulated by applying its spectral response to the 224-band hyperspectral images $\boldsymbol{x}_h$ of 10-m GSD (see [33, Sec. IV-A]), consequently yielding the 12-band ground truth $\boldsymbol{x}$.

Next, we describe how to generate the simulated Sentinel-2 image $\boldsymbol{y}$ from $\boldsymbol{x}$, where $\boldsymbol{y}$ will be fed into the algorithms under test. According to (1), the ground-truth images $\boldsymbol{x}$ are spatially blurred by 13×13 Gaussian kernels with variance set according to that estimated from ESA's data quality report on Sentinel-2 satellite products [51], followed by spatial downsampling with a factor of 1 (for B2–B4 and B8), 2 (for B5–B7, B8a, B11, and B12), and 6 (for B1 and B9) to simulate the GSD of real Sentinel-2 data (see Table I). Then, identically and independent distributed (i.i.d.) zero-mean Gaussian noise (with variance σ^2) of 40-dB SNR, defined as $\|\boldsymbol{y}\|_2^2/(\sigma^2 \ell n)$ (resulting in $\sigma = 0.6548$ for crop data, $\sigma = 1.5157$ for coastal data, $\sigma = 1.6352$ for mountainous data, and $\sigma = 3.0777$ for urban data), is added just to account for modeling error in (1). Finally, the synthesized 12-band multiresolution data, with bands of finest spatial resolution (i.e., 10-m bands) containing 432×108 pixels, are processed by Sentinel-2 super-resolution methods, including MUSA, ATPRK, and the proposed SSSS algorithm.

C. Quantitative Comparison Using Simulated Sentinel-2 Data

To quantitatively compare the proposed SSSS algorithm (i.e., Algorithm 1) with the state-of-the-art Sentinel-2 super-resolution methods, i.e., MUSA [33], ATPRK [22], DSen2 [17] and S2Sharp [24], we comprehensively adopt four commonly used quality metrics as defined next. The first metric, for measuring the spectral distortion, is the SAM [53], which is given in degrees and defined as

$$\mathrm{SAM} = \frac{1}{n}\sum_{i=1}^{n} \arccos\left(\frac{([\widehat{\boldsymbol{X}}]_{:,i})^T [\boldsymbol{X}]_{:,i}}{\|[\widehat{\boldsymbol{X}}]_{:,i}\|_2 \cdot \|[\boldsymbol{X}]_{:,i}\|_2}\right)$$

where $[\boldsymbol{X}]_{:,i}$ denotes the ith pixel of the target image (i.e., the ith column of $\boldsymbol{X}$) and $[\widehat{\boldsymbol{X}}]_{:,i}$ is its estimate. The second

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TABLE II

PERFORMANCE MEASURES ON THE CROP, COASTAL, MOUNTAINOUS, AND URBAN DATA SETS, WHERE THE BEST PERFORMANCE OF EACH MEASURE FOR EACH DATA SET IS INDICATED BY BOLDFACED NUMBER

Datasets	Methods	SAM (degree)	RMSE	SRE (dB)	UIQI	T (sec.)
Crop Dataset	MUSA	0.9518	55.5026	32.6243	0.9853	501.13
	ATPRK	1.2341	71.8985	31.5727	0.9819	226.32
	DSen2	2.4042	139.5301	28.2986	0.9442	**10.08**
	S2Sharp	1.9193	152.2005	26.9332	0.9530	20.78
	SSSS	**0.8040**	**50.0046**	**33.1973**	**0.9864**	100.05
Coastal Dataset	MUSA	1.8156	24.7417	27.9673	**0.9377**	455.17
	ATPRK	1.8678	21.1794	28.9295	0.9214	222.48
	DSen2	4.5553	57.5471	18.5766	0.9161	**10.23**
	S2Sharp	2.3934	47.7705	18.3089	0.9191	20.42
	SSSS	**1.1622**	**18.0857**	**29.3141**	0.9317	105.83
Mountainous Dataset	MUSA	1.0779	17.6223	31.3936	0.9915	400.62
	ATPRK	1.7754	25.6363	29.0110	0.9878	226.90
	DSen2	2.7074	48.3057	19.1132	0.9577	**10.20**
	S2Sharp	2.4254	51.5264	19.0960	0.9401	22.46
	SSSS	**1.0478**	**17.1165**	**31.6545**	**0.9916**	97.99
Urban Dataset	MUSA	1.5680	54.8530	27.2624	0.9727	425.72
	ATPRK	1.6179	47.1817	28.5205	**0.9885**	222.04
	DSen2	2.8366	108.9142	22.3203	0.9371	**9.77**
	S2Sharp	2.5630	102.3011	22.3814	0.9528	24.13
	SSSS	**1.0553**	**35.4103**	**30.0981**	0.9862	101.72

metric is the root-mean-square error (RMSE), which is defined as

$$\text{RMSE} = \frac{1}{\sqrt{\ell n}} \|\widehat{\boldsymbol{X}} - \boldsymbol{X}\|.$$

The third metric is the signal-to-reconstruction error (SRE) [32], given in dB, which is defined for each band b as

$$\text{SRE}_b = 10 \log_{10} \frac{\|\boldsymbol{x}_b\|_2^2}{\|\widehat{\boldsymbol{x}}_b - \boldsymbol{x}_b\|_2^2}$$

where $\widehat{\boldsymbol{x}}_b$ is the bth band of the super-resolved image. Finally, we also consider the universal image quality index (UIQI), defined for each band b as

$$\text{UIQI}_b = \mathbb{E}\left[\frac{\sigma^2_{\boldsymbol{x}\hat{\boldsymbol{x}}}}{\sigma_{\boldsymbol{x}}\sigma_{\hat{\boldsymbol{x}}}} \cdot \frac{2\mu_{\boldsymbol{x}}\mu_{\hat{\boldsymbol{x}}}}{\mu^2_{\boldsymbol{x}} + \mu^2_{\hat{\boldsymbol{x}}}} \cdot \frac{2\sigma_{\boldsymbol{x}}\sigma_{\hat{\boldsymbol{x}}}}{\sigma^2_{\boldsymbol{x}} + \sigma^2_{\hat{\boldsymbol{x}}}}\right]$$

where $\boldsymbol{x}$ is the ground-truth image in a 64×64 sliding window at the given band b and $\hat{\boldsymbol{x}}$ is its estimate, $\mathbb{E}$ is an averaging operator over windows, $\sigma^2_{\boldsymbol{x}\hat{\boldsymbol{x}}}$ is the covariance between $\boldsymbol{x}$ and $\hat{\boldsymbol{x}}$, $\sigma_{\boldsymbol{x}}$ and $\sigma_{\hat{\boldsymbol{x}}}$ are the standard deviations of $\boldsymbol{x}$ and $\hat{\boldsymbol{x}}$, respectively, and $\mu_{\boldsymbol{x}}$ and $\mu_{\hat{\boldsymbol{x}}}$ are the means of $\boldsymbol{x}$ and $\hat{\boldsymbol{x}}$, respectively [54]. The three terms in UIQI account for the loss of correlation, luminance distortion, and contrast distortion. The range of UIQI is $[-1, 1]$, and its upper bound is achieved if and only if $\widehat{\boldsymbol{X}} = \boldsymbol{X}$.

The results are summarized in Table II, where one can see that the proposed SSSS algorithm has uniformly best performance. Specifically, as indicated by SAM, SSSS yields significantly much better spectrum estimate, especially for the coastal and urban data sets. It is interesting to notice that MUSA also preserves good spectral property for the crop and mountainous data sets mainly due to its low-rank approximation of the target image. However, ATPRK sharpens each band separately, making the spectral distortion in the super-resolved image slightly higher. In terms of RMSE, a global quality measure, MUSA, has better performance than ATPRK on the crop and mountainous data sets, while the performance of SSSS is still best for all the data sets. Generally, unsupervised methods perform better, while, as will be seen soon, the supervised method DSen2 is much faster.

As for SRE and UIQI, their values for each band are plotted as curves in Figs. 7 and 8, respectively. One would see that for the 20-m bands, the two self-similarity-guided methods (i.e., SSSS and MUSA) have a similar performance and yield much better results than the kriging-based method (i.e., ATPRK). We also notice that for the 60-m bands, ATPRK has the best performance on B1 (except for the mountainous data set), while SSSS performs well on B9 (except for the crop data set). We remark that for some tough scenarios (e.g., the super-resolution of 60-m bands), the CNN-based DSen2 performs above average, such as when super-resolving B1 of the urban data set, as shown in Figs. 7 and 8. This shows the potential of CNN, especially when the amount of images acquired by Sentinel-2 (a new satellite) becomes comparable to that of conventional RGB images in the future. On the other hand, S2Sharp occasionally achieves the best performance (e.g., B5 of the crop area), as shown in Fig. 8. For each data set, the averaged values of SRE and UIQI (averaged over spectral bands) are given in Table II, where one can see that the proposed SSSS algorithm has the best performance.

We remark that we are not proving that SSSS is perfect; in fact, there is no perfect algorithm. For some very tough scenarios, e.g., super-resolving B11 and B12 (far from all 10-m bands) of urban data (with complicated texture), SSSS still cannot outperform the benchmark method ATPRK, which performs best here (see Figs. 7 and 8). What we actually aim to prove is the high potential and effectiveness

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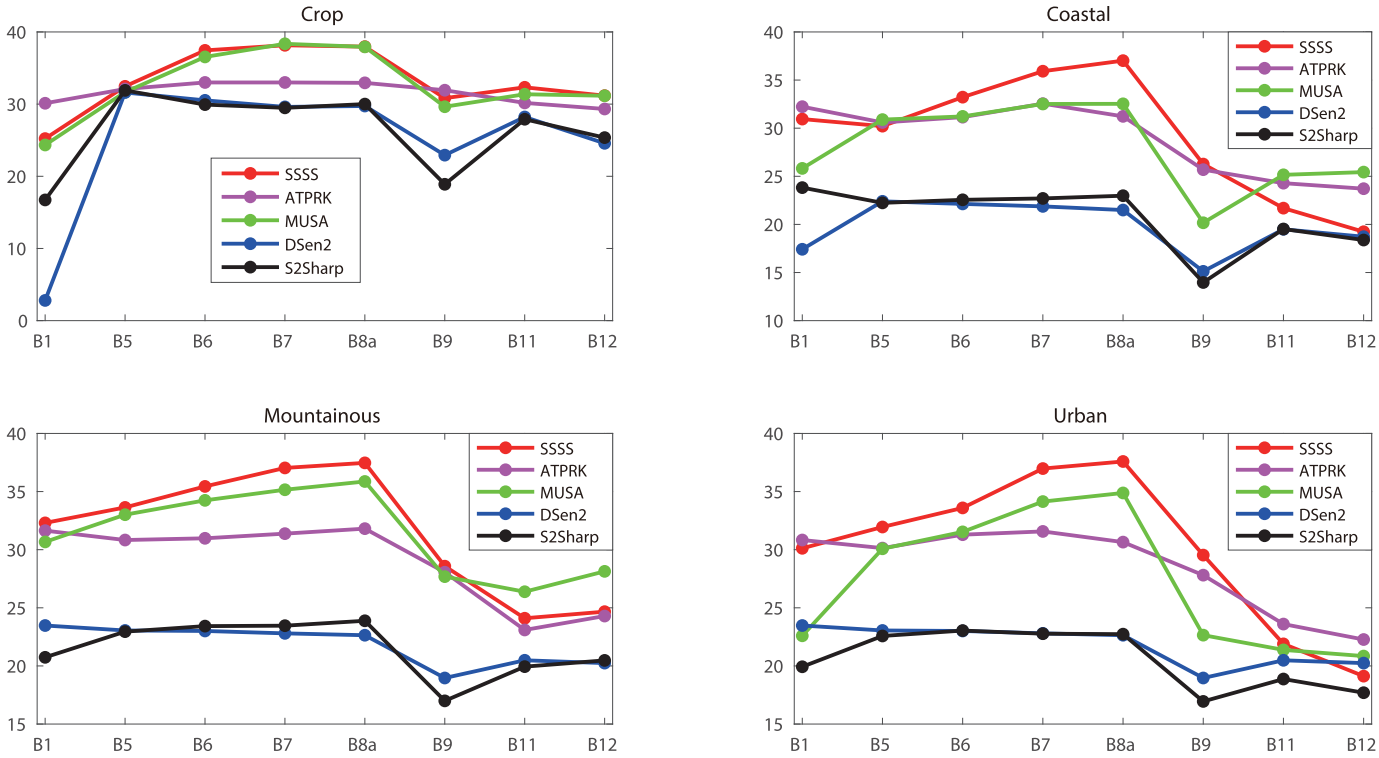


Fig. 7. SRE values (dB) of various supervised/unsupervised Sentinel-2 super-resolution methods tested on the crop, coastal, mountainous, and urban data sets. The larger the value of SRE, the better the super-resolution performance.

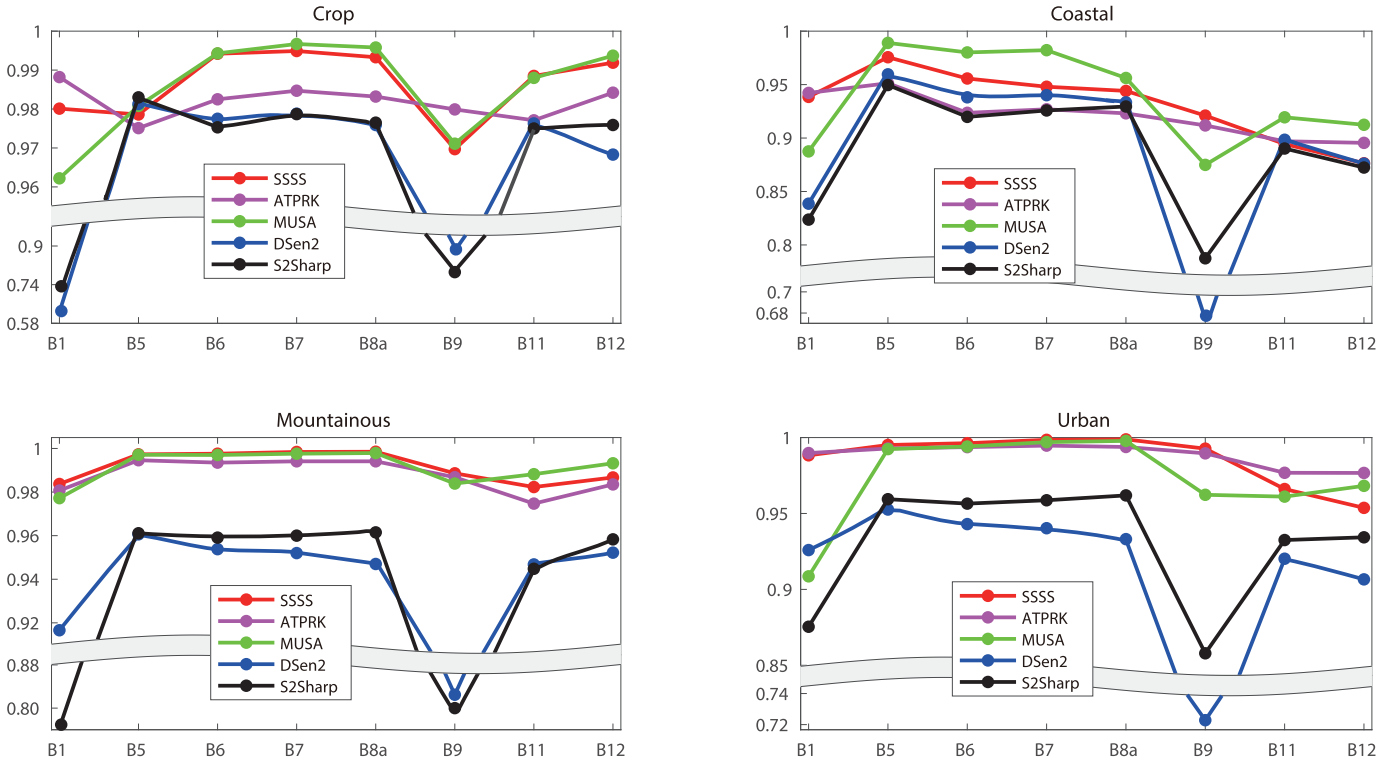


Fig. 8. UIQI values of various supervised/unsupervised Sentinel-2 super-resolution methods tested on the crop, coastal, mountainous, and urban data sets. The value of UIQI is equal to 1 if, and only if, the target image is perfectly reconstructed.

of bringing the self-similarity regularization into the convex framework, with which the strength of self-similarity prior would be better exploited from a computational point of view (e.g., convergence guarantee; see Theorem 1).

Finally, we show the computational time in Table II. All the experiments mentioned earlier are executed on a computer facility equipped with Core-i5-7200U CPU with 2.50-GHz speed and 8-GB RAM, and all the methods are implemented

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in Mathworks MATLAB R2019a. One can see that the CNN-based DSen2 is the fastest one (about one order of magnitude faster than SSSS). Among the unsupervised methods, S2Sharp is fastest, while the proposed SSSS algorithm is second fastest. SSSS is about four-to-five times faster than MUSA, which is also based on implicit self-similarity regularization and relies on plug-and-play strategy, alluding that the explicitly defined self-similarity regularizer in SSSS does help reduce the computational burden while yielding better super-resolution performance.

V. Conclusion and Future Work

In this article, from the Sentinel-2 images, we adaptively learn a self-similarity graph that is used to explicitly define the self-similarity phenomenon (widely observed in natural images) as a convex regularizer and incorporate it into a convex formulation for the Sentinel-2 super-resolution problem. We develop an algorithm, termed SSSS, to solve this problem based on SALSA (or ADMM) with provable convergence guarantee, where computationally efficient closed-form solutions for the involved subproblems are derived. Experiments are conducted on real and synthetic Sentinel-2 images, showing the great superiority of SSSS over the existing benchmark methods.

Just like there are several variations of TV regularization, it is very interesting to explore other variations of the self-similarity regularization while still keeping the convexity. This would help upgrade the performance of Sentinel-2 super-resolution in some very tough scenarios and would have a significant impact when introducing our new convex self-similarity framework in regularizing other imaging inverse problems. Specifically, one potential line of attack is to exploit the flexibility of our definition of ϕ, as discussed in Section III (see Fact 1), which will be left for future work. Though the default setting of $(\lambda, \mu) = (0.1, 0.1)$ in SSSS works well for several geographical areas on the Earth surface, exploring automatic parameter selection (e.g., using GCV [24]) is another interesting future work. The major objective of this article lies within the regularization design, i.e., bringing the important self-similarity regularization into the convex framework, whose potential usage in other imaging inverse problems can be anticipated.

Appendix A
Proof of Theorem 1

Based on a more general result by Eckstein and Bertsekas [55, Th. 8], the convergence of SALSA has been proven in [42] under the following mild assumptions, all satisfied in our case: 1) the terms of objective function are closed, proper, and convex; 2) the optimization has a solution; 3) the null space of the linear map between the original variables (i.e., $\boldsymbol{z}$ in our case) and the constraint variables (i.e., $(\boldsymbol{v}, \{\boldsymbol{V}_{i,j}\})$ in our case) is just the zero vector; and 4) $\mu > 0$.

Assumption 1) directly follows from the premise of $\alpha_{i,j} \geq 0$, $\forall (i,j) \in \mathcal{K} \subseteq \{(i,j) \mid i,j \in \mathcal{I}_N,\ i \neq j\}$, and the fact that the composition of a convex function (i.e., $\|\cdot\|_F^2$) and an affine function (i.e., $(\boldsymbol{P}_i - \boldsymbol{P}_j)\boldsymbol{V}_{i,j}$) preserves convexity [45]. Assumptions 2) and 4) are obvious. As for assumption 3), we recall that $\boldsymbol{U}$ has full column rank, which, together with some basic property of Kronecker product [i.e., $\operatorname{rank}(\boldsymbol{A}\otimes\boldsymbol{B}) = \operatorname{rank}(\boldsymbol{A})\operatorname{rank}(\boldsymbol{B})$], implies that $\boldsymbol{U}\otimes\boldsymbol{I}_n$ is also of full column rank; therefore, $\boldsymbol{v} = (\boldsymbol{U}\otimes\boldsymbol{I}_n)\boldsymbol{z} = \boldsymbol{0}$ implies $\boldsymbol{z} = \boldsymbol{0}$. Thus, all the assumptions hold true, and Theorem 1 has been proven. ■

Appendix B
Proof of (14)

Since the target image $\boldsymbol{X}$ contains $n_1' \times n_2'$ pixels, $\boldsymbol{B}_b$ is a BCCB matrix having $n_2' \times n_2'$ circulant blocks with each block being an $n_1' \times n_1'$ circulant matrix [49]. From basic matrix analysis, one can show that $(\boldsymbol{F}_{n_2'}\otimes\boldsymbol{F}_{n_1'})\boldsymbol{B}_b(\boldsymbol{F}_{n_2'}\otimes\boldsymbol{F}_{n_1'})^H$ is diagonal. As $(\boldsymbol{F}_{n_2'}\otimes\boldsymbol{F}_{n_1'})^{-1} = (\boldsymbol{F}_{n_2'}\otimes\boldsymbol{F}_{n_1'})^H$, there exists a diagonal matrix $\boldsymbol{\Sigma}$ such that

$$\boldsymbol{B}_b = (\boldsymbol{F}_{n_2'}\otimes\boldsymbol{F}_{n_1'})^H\boldsymbol{\Sigma}(\boldsymbol{F}_{n_2'}\otimes\boldsymbol{F}_{n_1'}) \tag{20}$$

where the diagonal vector of $\boldsymbol{\Sigma}$ is the vectorized version of the 2-D DFT of the $n_1' \times n_2'$ convolution kernel (or blurring kernel). Also, the downsampling matrix can be explicitly written as

$$\boldsymbol{M}_b = \boldsymbol{M}_2^T \otimes \boldsymbol{M}_1 \tag{21}$$

where $\boldsymbol{M}_1 \triangleq \boldsymbol{I}_{n_1'/r_i}\otimes(\boldsymbol{e}_1^{(r_i)}(\boldsymbol{e}_1^{(r_i)})^T) \in \mathbb{R}^{n_1'\times n_1'}$, $\boldsymbol{M}_2 \triangleq \boldsymbol{I}_{n_2'/r_i}\otimes(\boldsymbol{e}_1^{(r_i)}(\boldsymbol{e}_1^{(r_i)})^T) \in \mathbb{R}^{n_2'\times n_2'}$, and r_i is the upsampling factor of band b. We need Proposition 1, whose proof follows from [49, eq. (9)] and the observation that both $\boldsymbol{M}_1$ and $\boldsymbol{M}_2$ are diagonal with all the diagonal elements being either 0 or 1.

Proposition 1: Let $\boldsymbol{M}' \triangleq \boldsymbol{I}_{n'/r}\otimes(\boldsymbol{e}_1^{(r)}(\boldsymbol{e}_1^{(r)})^T) \in \mathbb{R}^{n'\times n'}$, and $\boldsymbol{F}_{n'} \in \mathbb{C}^{n'\times n'}$ be the DFT matrix. Then

$$\boldsymbol{F}_{n'}\boldsymbol{M}'\boldsymbol{F}_{n'}^H = \boldsymbol{F}_{n'}^H\boldsymbol{M}'\boldsymbol{F}_{n'} = \frac{1}{r}(\boldsymbol{1}_r\otimes\boldsymbol{I}_{n'/r})\big(\boldsymbol{1}_r^T\otimes\boldsymbol{I}_{n'/r}\big).$$

On the other hand, we have $\boldsymbol{M}_b = \boldsymbol{M}_2\otimes\boldsymbol{M}_1$ and $\boldsymbol{M}_b^T\boldsymbol{M}_b = \boldsymbol{M}_b\boldsymbol{M}_b = \boldsymbol{M}_b$.

For notational conciseness, we define $\boldsymbol{F} \triangleq \boldsymbol{F}_{n_2'}\otimes\boldsymbol{F}_{n_1'} \in \mathbb{C}^{n\times n}$ and $\boldsymbol{D} \triangleq (\boldsymbol{1}_{r_i}\otimes\boldsymbol{I}_{n_2'/r_i})\otimes(\boldsymbol{1}_{r_i}\otimes\boldsymbol{I}_{n_1'/r_i}) \in \mathbb{R}^{n\times n_i}$. Then, combining (20), (21), Proposition 1, and a basic property of Kronecker product (i.e., for any matrices $\boldsymbol{A}'$, $\boldsymbol{B}'$, $\boldsymbol{C}'$, and $\boldsymbol{D}'$ of proper dimensionality, $(\boldsymbol{A}'\otimes\boldsymbol{B}')(\boldsymbol{C}'\otimes\boldsymbol{D}') = (\boldsymbol{A}'\boldsymbol{C}')\otimes(\boldsymbol{B}'\boldsymbol{D}')$), we arrive at

$$\begin{aligned}\boldsymbol{B}_b^T\boldsymbol{M}_b^T\boldsymbol{M}_b\boldsymbol{B}_b &= \boldsymbol{F}^H\boldsymbol{\Sigma}^H\boldsymbol{F}(\boldsymbol{M}_2\otimes\boldsymbol{M}_1)\boldsymbol{F}^H\boldsymbol{\Sigma}\boldsymbol{F}\\ &= \boldsymbol{F}^H\boldsymbol{\Sigma}^H\big(\boldsymbol{F}_{n_2'}\boldsymbol{M}_2\boldsymbol{F}_{n_2'}^H\otimes\boldsymbol{F}_{n_1'}\boldsymbol{M}_1\boldsymbol{F}_{n_1'}^H\big)\boldsymbol{\Sigma}\boldsymbol{F}\\ &= \boldsymbol{F}^H\boldsymbol{\Sigma}^H\boldsymbol{D}\frac{1}{r_i^2}\boldsymbol{D}^H\boldsymbol{\Sigma}\boldsymbol{F}\end{aligned} \tag{22}$$

which, together with the Woodbury matrix identity [45] and $\boldsymbol{F}\boldsymbol{F}^H = \boldsymbol{I}_n$, yields

$$\begin{aligned}&\big(\boldsymbol{B}_b^T\boldsymbol{M}_b^T\boldsymbol{M}_b\boldsymbol{B}_b + \mu\boldsymbol{I}_n\big)^{-1}\\ &= \left(\boldsymbol{F}^H\boldsymbol{\Sigma}^H\boldsymbol{D}\frac{1}{r_i^2}\boldsymbol{D}^H\boldsymbol{\Sigma}\boldsymbol{F} + \mu\boldsymbol{I}_n\right)^{-1}\\ &= \frac{1}{\mu}\boldsymbol{I}_n - \frac{1}{\mu^2}\boldsymbol{F}^H\boldsymbol{\Sigma}^H\boldsymbol{D}\left(r_i^2\boldsymbol{I}_{n_i} + \frac{1}{\mu}\boldsymbol{D}^H|\boldsymbol{\Sigma}^2|\boldsymbol{D}\right)^{-1}\boldsymbol{D}^H\boldsymbol{\Sigma}\boldsymbol{F}.\end{aligned}$$

Thus, the proof of (14) is completed. ■

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APPENDIX C
PROOF OF THE DIAGONALITY IN (14)

In this appendix, we prove that $(r^2\boldsymbol{I}_{n_i} + (1/\mu)\boldsymbol{D}^H|\boldsymbol{\Sigma}^2|\boldsymbol{D})$ is diagonal. We need Fact 2.

Fact 2: Any two matrices $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ (of appropriate dimension) satisfy the following.

1) If each row of $\boldsymbol{A}_k$ ($k \in \{1, 2\}$) has maximally one nonzero entry, then each row of $\boldsymbol{A}_1 \otimes \boldsymbol{A}_2$ also has maximally one nonzero entry.
2) If $\boldsymbol{A}_1$ is diagonal, and each row of $\boldsymbol{A}_2$ has maximally one nonzero entry, then $\boldsymbol{A}_2^H\boldsymbol{A}_1\boldsymbol{A}_2$ must be diagonal.

The proof of Fact 2 simply follows from the definitions of Kronecker product and matrix multiplication. Since each row of $\mathbf{1}_{r_i} \otimes \boldsymbol{I}_{n_k/r_i}$ ($k \in \{1, 2\}$) has maximally one nonzero entry, Fact 2-1) tells us that each row of $\boldsymbol{D} \triangleq (\mathbf{1}_{r_i} \otimes \boldsymbol{I}_{n_2/r_i}) \otimes (\mathbf{1}_{r_i} \otimes \boldsymbol{I}_{n_1/r_i}) \in \mathbb{R}^{n\times n_i}$ has maximally one nonzero entry. Then, Fact 2-2) implies that $\boldsymbol{D}^H|\boldsymbol{\Sigma}^2|\boldsymbol{D}$ is diagonal. Thus, $(r^2\boldsymbol{I}_{n_i} + (1/\mu)\boldsymbol{D}^H|\boldsymbol{\Sigma}^2|\boldsymbol{D})$ is also diagonal. ■

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